

Automatic Speech Recognition for Low-Resource Languages: A Survey of Self-Supervised and Cross-Lingual Representation Learning

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Abstract—Automatic Speech Recognition (ASR) for low-resource languages is an extremely challenging research area due to the scarcity of annotated data collections, strong dialectal diversity, and natural acoustic variability. The advent of self-supervised learning (SSL) has fundamentally changed the ASR paradigm by enabling representation learning from large-scale unlabeled audio data. This survey examines the current state of SSL-based ASR for low-resource languages, including foundational pre-training models (Wav2Vec 2.0, HuBERT, WavLM), cross-lingual and multilingual transfer methods (XLSR, XLS-R, IndicWav2Vec), fine-tuning and data-efficient techniques, and architectural choices between Connectionist Temporal Classification (CTC) and encoder-decoder models. Empirical case studies spanning Telugu, Swedish, Maltese, and several African languages are discussed. Key findings indicate that SSL significantly reduces the requirement for labeled data; however, word error rates for morphologically complex languages remain high, and standardized benchmarking protocols for low-resource settings are notably lacking. Research gaps and future directions are outlined.

Index Terms—automatic speech recognition, low-resource languages, self-supervised learning, cross-lingual transfer, Wav2Vec 2.0, HuBERT, Whisper, Indic languages, fine-tuning, word error rate

I. INTRODUCTION

Low-resource languages represent a fundamental bottleneck in Automatic Speech Recognition (ASR) deployment because traditional supervised pipelines based on Hidden Markov Model–Gaussian Mixture Model (HMM-GMM) or Deep Neural Network–Hidden Markov Model (DNN-HMM) architectures require large corpora of manually labelled data and language-specific phonemic lexicons, which are often unavailable or prohibitively expensive to construct for underrepresented languages [1]. Consequently, thousands of the world’s languages remain inaccessible to ASR technology.

Self-supervised learning (SSL) has begun to address this imbalance by leveraging large-scale unlabelled audio to learn rich acoustic representations, thereby drastically reducing the need for annotated data [2]. SSL-pretrained models can be fine-tuned using only minutes of transcribed speech, opening a viable pathway for ASR deployment in linguistically excluded communities.

This survey takes a comprehensive look at the current state-of-the-art concerning self supervised learning (SSL) based automatic speech recognition (ASR) for low resource languages, considering 5 thematic axes: (1) SSL pre-training paradigms; (2) cross-lingual and multi-lingual transfer strategies; (3) fine-tuning and efficient data adaptation strategies; (4) architectural comparisons between CTC based and encoder-decoder models; (5) applied case studies of low resource ASR with special focus on Indic languages such as Telugu [1].

The survey collates pragmatic insights, summarises the empirical findings and highlights research gaps that need to be addressed if robust and real-world deployment is to be assured.

II. FUNDAMENTALS OF SELF-SUPERVISED SPEECH REPRESENTATION LEARNING

The rationale behind SSL for speech is that raw audio signals contain rich acoustic and phonemic patterns that can be learned without manual annotation. A landmark model, **Wav2Vec 2.0** [2], introduced by Baevski et al., employs a contrastive learning objective over masked latent representations. Its architecture consists of a multi-layer convolutional feature encoder that extracts frame-level latent features from raw waveforms, followed by a Transformer-based context network that captures long-range temporal dependencies. During pretraining, a portion of the latent sequence is masked, and the model is trained to identify the correct representation from a set of distractor samples drawn from a learned quantised codebook. Fine-tuning with Connectionist Temporal Classification (CTC) [2] enables efficient adaptation to transcription tasks. Notably, Wav2Vec 2.0 achieves state-of-the-art performance on LibriSpeech using as little as 10 minutes of labelled data.

Hsu et al. proposed **HuBERT** [3], which improves the SSL framework by predicting masked hidden units derived from offline clustering of acoustic features (e.g., MFCC-based k -means assignments). In spite of the traditional contrastive objective used by Wav2Vec 2.0, HuBERT uses cross-entropy loss through pseudo-labels generated through clustering followed by relabelling. The training regime of this architectural choice is more stable, and this plus the sensitivity of the model

to initial quality of clustering will allow HuBERT to reach competitive with Wav2Vec 2.0 Word Error Rates, often higher, when trained on comparable supervision budgets, especially in noisy and spontaneous speech settings.

Chen et al. extended this framework with **WavLM** [9], which combines the HuBERT masked prediction objective with a speech denoising task. By training jointly on clean and artificially corrupted speech, WavLM learns representations that are both discriminative and acoustically robust. WavLM achieves state-of-the-art performance on the SUPERB benchmark [11] across ten speech processing tasks, including ASR, speaker verification, and emotion recognition.

Mohamed et al. [12] are a careful and systematic overview of self-supervised speech representation learning, iterating more than ten years of approach improvement in contrastive, predictive, and generative paradigms. The evaluation of SSL models has been standardized with the SUPERB benchmark [11], which is based on a unified evaluation suite, which makes use of frozen upstream models, which are supplemented with a light-weight downstream heads. The adoption of a SUPERB-like evaluation framework in low-resource contexts is still an urgent problem that requires solutions.

Strengths and Limitations

The SSL models significantly reduce the need for manual annotation of data, making use of Transformer architectures for long-range contextual phonemic modelling. Nonetheless, pre-training requires a large amount of computational resources, that is, large GPU clusters and long training times. Performance degrades if there is a significant difference between the phonological inventory of the target language and that of the pre-training data distribution, and if there may not be a sufficiently large unlabeled corpus available for endangered languages.

III. CROSS-LINGUAL AND MULTILINGUAL TRANSFER LEARNING

Initial models of SSL were mostly monolingual, and so limited their relevance to low-resource languages. Conneau et al. introduced XLSR [4], which is pre-trained on 56 languages with the Wav2Vec 2.0 contrastive objective. Their central hypothesis is that universal structures of phonetics and prosody are found in all languages of the world, and that the structure of a common parameter space can be drawn from multilingual data to account for these regularities. XLSR obtained substantial improvements in Phoneme Error Rate (PER) and Word Error Rate (WER) of low-resource languages, including Swahili and Kyrgyz when fine-tuned using scarce annotated data, validating what can only be termed the cross-lingual transfer hypothesis [4].

Babu et al. further refined this with XLS-R [5], which was pre-trained on 436,000 hours of the speech of 128 languages. The XLS-R-300M variant is a pragmatic compromise between multi language coverage and computation feasibility. Evaluations on CoVoST - 2 and VoxPopuli had

shown the increased robustness against noise, accent variation and dialectal diversity compared to monolingual baselines [5].

Regarding Indic languages, Javed et al. developed IndicWav2Vec [10], pre-training on 40 Indian languages with at around 17,000 hours of speech. IndicWav2Vec retains some unique phonological elements in the group of Indic languages such as retroflex consonants, aspirated stops and vowel length contrast which are less available in globally pre-trained models. Fine tuning IndicWav2Vec on Telugu leads to significantly lower WER compared to just fine tuning XLS-R, which again highlights the utility of language family specific pre-training as a complementary approach.

Yadav and Sitaram [13] give an in-depth review of multilingual ASR models, which also focuses on the effectiveness of cross-lingual transfer, which is very much dependent on the phonological and typological similarity between the pre-training language and the target language. This is particularly relevant for Dravidian languages like Telugu which there is limited overlap with the Indo-Euro languages used in the majority of pre-training multilingual corpora.

Strengths and Limitations

Multilingual pre- ability is good cross-linguistic transfer of phonetics, it reinforces resistance to noise, accent and dialect. However, parameter counts and training time scale with the number of languages; loss of transfer efficiency is reduced when the languages are typologically distant, and unbiased corpus representation attacks minority language families surprisingly.

IV. FINE-TUNING AND DATA-EFFICIENT ADAPTATION STRATEGIES

The successful adaptation of pretrained self-supervised learning (SSL) functions to low resource target languages relies on the meansly crafted fine-tuning pipelines. In particular, the calibre, variety and preprocessing of fine-tuning data, the choice of augmentation techniques, and the decoding strategy have profound impact on the downstream automatic speech recognition (ASR) performance.

Park and colleagues proposed SpecAugment [14], the computationally light weight data augmentation method, which directly applies frequency masking, time masking, and temporal warping operations to log -mel spectrograms during training. This type of approach has quickly made itself a staple in today's ASR fine tuning, due to its simplicity and the consistent and strong empirical gains it provides in low data regimes.

In a latest study, Bharadwaj et al. [1] fine tuner the XLS-R-300M model in Telugu Microsoft Speech Corpus which gave result with 42.98% of WER and 9.65% of CER. Given the large WERcary difference, the well documented difficulty in morphologically rich languages (Agglutination languages such as Telugu) in which multiple morphemic units could occur at once within the same lexical item and the word boundaries do not have a clear phonetic demarcation. This work also introduced a real time, Flask based web interface

for model inference, this represents one of the few end-to-end deployments of an SSL based Indic ASR system.

Chadha et al. developed the **Vakyansh** toolkit [15], an open-source end-to-end ASR framework supporting 18 Indic languages, integrating data collection, preprocessing, Wav2Vec 2.0 fine-tuning, and KenLM n -gram language model decoding. Incorporating a language model during beam-search decoding substantially reduces WER relative to greedy CTC decoding alone, highlighting decoding strategy as a frequently underexplored optimisation lever.

Al-Ghezi et al. [6] presented the research of self-supervised learning (SSL)-based automatic speech recognition (ASR) systems based on the target user namely second language (L2) Swedish speakers that resulted in significant reductions in term error rates (WER) from traditional hidden Markov model (HMM)-deep neural network (DNN) baselines. The results suggest that SSL obtained representations capture speaker independent acoustic features that are capable of generalization across divergent nonnative phonological realizations.

A phenomenon evident in these investigations, and many other investigations, is that fine-tuning SSL models using small quantities of labeled data result in salient WER improvements. However, performance breaks down disproportionately in contexts where they are confronted with morphologically complex lexical forms, code switched utterances and dialectal variants which fall outside the fine tuning distribution. These results provide a case for more aggressive exploration of dialect-aware data augmentation, language models, and parameter-efficient adapter-based fine-tuning, and popularizing them as complementary approaches.

V. COMPARATIVE ANALYSIS OF SSL-BASED ASR ARCHITECTURES

Modern SSL-based ASR systems can be broadly categorized into two main architectural families, each with distinct trade-offs.

A. CTC-Based Transformer Models

The Wav2Vec 2.0 [2], HuBERT [3], WavLM [9], XLSR [4], XLS-R [5], and IndicWav2Vec [10] all follow a three stage pipeline that starts from a convolutional feature encoder, followed by a Transformer-based context encoder and finally a Connectionist Temporal Classification (CTC) output layer. CTC decoding is a truly non-autoregressive algorithm which will conduct parallel computations over temporal indices, whereas this enables low latency inference which makes these models very suitable for real-time applications and edge device applications. Indeed they easily mesh together with conventional beam-search CTC decoders and n -gram language models.

The main drawback of the CTC approach is its assumption of conditional independence between frames of output which makes it difficult to do internal language modeling (therefore can result into higher Word Error Rates, i.e. WERs) when facing linguistically complex sequences.

B. Encoder-Decoder Transformer Models

Radford et al. proposed **Whisper** [7], a full sequence-to-sequence Transformer that was trained on a tremendous amount of audio data (680,000 hours) from the Web, and weakly supervised. First, log-mel spectrograms are passed through a Transformer encoder and then a Transformer decoder autoregressively produces the transcription. This decoder has a builtin tendency to incorporate the modeling of language, providing contextually sensible transcripts that is particularly useful for continuous speech which will be filled with a complicated syntax.

Williams et al. [8] made a direct empirical comparison of Wav2Vec 2.0 with Whisper on Maltese ASR - a morphologically rich and low-resource Semitic language. However, Whisper became stronger under real-world noisy conditions, albeit with substantially higher per-inference latency (3-5x). Such findings are in line with theoretical expectations: autoregressive decoding makes use of contextual information from previously-generated tokens at the cost of computation over time.

C. Summary Comparison

Table I summarises the key SSL-based ASR models reviewed, their architectures, training paradigms, and primary strengths.

TABLE I
COMPARISON OF SSL-BASED ASR MODELS FOR LOW-RESOURCE LANGUAGES

Model	Architecture	Multi-lingual	Key Strength
Wav2Vec 2.0 [2]	CNN + Transformer + CTC	Limited	High label efficiency
HuBERT [3]	CNN + Transformer + CTC	Yes	Stable pretraining
WavLM [9]	CNN + Transformer + CTC	Yes	Noise robustness; SUPERB SOTA
XLSR [4]	Multilingual Transformer + CTC	56 langs	Strong cross-lingual transfer
XLS-R [5]	Scaled Multilingual + CTC	128 langs	Robust multilingual modeling
IndicWav2Vec [10]	Multilingual Transformer + CTC	40 Indic	Indic phonetic adaptation
Whisper [7]	Transformer Encoder-Decoder	Massive	Noise robustness; LM integration

In the context of the deployment of Automatic Speech Recognition system with limited computational resources, CTC models, coupled with n -gram language model decoding

offers the most efficient tradeoff between accuracy of transcription and operational feasibility; while encoder-decoder models are the method of choice when it is possible to afford the extra latency incurred by the performance of inference on the server side.

VI. APPLIED LOW-RESOURCE ASR CASE STUDIES

Applied investigations serve as a way of realizing the theoretical potential of self supervised learning based automatic speech recognition in real world observable results in order to elucidate the practical problems associated with deployment.

The Telugu ASR system pioneered by Bharadwaj et al. [1] is one of the most end-to-end case studies that has been carried out on a Dravidian language which is low resource. Having fine tuned the XLS-R-300M architecture on the Microsoft speech corpus and mounted the model-based results using Flask (a Python framework similar to Java's Spring) to build a real-time web-based interface, the authors have shown end-to-end feasibility. This pipeline is the sum of the multilingual SSL pre-training, step(s) of supervised domain-specific adaptation, and inference accessible to users without the technical expertise. The resulting Word Error Rate (WER) of 42.98% that is associated with a Character Error Rate (CER) of 9.65% emphasises the difficulties of word level instigated by agglutinative morphology and so gives cause for intervention in terms of, for example, n-gram language model decoding and morpheme-aware tokenisation.

Bhogale et al. proposed the **Vistaar** benchmark [16], a romanticized evaluation environment that contains twelve Indian languages in a variety of domains-various forms of speech, such as news broadcasting, conversational speech, and read speech. Vistaar enables domain-controlled, direct comparisons of ASR systems based on SSL in the Indic milieu. A salient observation which emerges from the evaluation is that the WER of conversational and spontaneous speech is four to eight times the WER of read speech for all the models evaluated. This is a crucial outcome for understanding that the mismatch between the domain of fine-tuning data and the conditions of deployment are often much more critical than choosing the model architecture in the first place.

Ogunremi et al. [17] studied SSL-based ASR in low-resource African languages with pervasive code-switching. By using multilingual SSL representations, they improved the transcription of code switched utterances using cross-lingual phonological correspondences. In turn, the combination of a code-switching awareness n-gram language model with the decoding model led to a WER reduction of approximately twenty percent compared to purely monolingual decoding. The implications of these results apply easily to Telugu and other Indic languages where English code-switching is common in the urban and semi-urban population.

Taken as a whole, the applied case studies in Telugu [1], Swedish [6], Maltese [8], and African languages [17] disclose three (unbroken) patterns. First, the dependency on annotated datasets is brought down significantly by SSL models. Second, cross lingual transfer shows greater fine tuning performance in any case where typological affinities exist between the pre training and target languages. Lastly, inconsistencies within the design of benchmark tests - test domain, speaker demographics and evaluation protocols - limits the interpretability and generality of the published WER metrics.

VII. DISCUSSION AND RESEARCH GAPS

The literature that has emerged derived concludes that there are five main trends of low resources SSL-based ASR:

- 1) Self-supervised learning has shifted the emphasis of ASR to representation-based learning, thus allowing it to be applied to languages with little associated annotated corpus.
- 2) Multilingual pre-training enables good cross-lingual phonetic transfer e.g., IndicWav2Vec has shown better transfer to Dravidian languages when compared with global pre-training models
- 3) Fine-tuning strategy is the major influence over downstream performance in that data augmentation, particularly SpecAugment [14], and integrating language models at the decoding stage requires joint optimisation.
- 4) Models based on CTC are more suitable for a real-time and edge deployment scenarios while encoder-decoders model allows contextual richer modelling for offline tasks requiring better accuracy.
- 5) There are not standardised and domain stratified benchmarks for low resource languages, forming a severe limitation to reproducibility and cross-study comparison.

Despite all these improvements, there is still a long way to go: the WER for dialect-rich and morphologically complex low-resource languages like Telugu still fluctuates around 40% and how to scale a research prototype into a robust, production-grade system is simply not detailed enough. Little systematic study has been done, however, concerning the synergy of fine tuning of data selection, decoding configurations and either scale of language model, or constraint on production-level of the latency. The consequences of code switching, dialectal variation and domain mismatch are loosely defined for the majority of the Indic languages as well, once deployed.

Future research should prioritise: (i) the development of morpheme-sensitive tokenisation schemes and architectures for agglutinative languages; (ii) dialect-sensitive and code-switching-resilient data augmentation; (iii) parameter-efficient fine-tuning paradigms e.g. adapter modules; and (iv) creation of standardised benchmarking pipelines reflecting deployment realities which is an extension of the Vistaar [16] and SUPERB [11] frameworks

VIII. CONCLUSION

This survey has surveyed the state of the art of self-supervised and cross-lingual representation learning for low-resource automatic speech recognition (ASR), including the basic pre-training schemes, multilingual transfer schemes, fine-tuning schemes, architectural trade-offs as well as empirical case studies.

Empirical data give numerous examples of self-supervised learning (SSL) greatly reducing the amount of annotated data necessary to build practical ASR systems, and thus greatly expanding the availability of this technology to a significantly larger number of the world's languages.

Despite these improvements, the general challenge of high-accuracy, dialect-robust, and code-switching-resilient ASR for morphologically rich languages still stands in a particularly acute form for Telugu because of the challenges encountered with agglutinative language structures in terms of accuracy, resilience to dialectal variation, code switched speech, and reproducibility of cross study evaluation.

Overcoming these shortcomings will require the development of model architecture design, strategies for data augmentation, decoding strategies and benchmark infrastructures.

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